

NLTX: Natural Language Transaction Exchange

Conversational Blockchain Payment System

A B.Tech Project Report Submitted in Partial Fulfillment of the Requirements
for the Award of the Degree of

Bachelor of Technology

Submitted by

Krishna Ramesh Bandewar

Registration No.: 2022BCS001

Department of Computer Science and Engineering

Under the Guidance of

Prof. A. M. Bainwad



Department of Computer Science and Engineering
Shri Guru Gobind Singhji Institute of Engineering and Technology
Nanded – 431606, Maharashtra, India

2026

Certificate

This is to certify that the report entitled “NLTX: Natural Language Transaction Exchange Conversational Blockchain Payment”, submitted by Krishna Bandewar 2022BCS001) to Shri Guru Gobind Singhji Institute of Engineering and Technology (SGGSIE&T), Nanded, for the award of the degree of **Bachelor of Technology in Computer Science and Engineering**, is a record of the original, bona fide work carried out by him under our supervision and guidance.

The work presented in this report has not been submitted in part or in full to any other University or Institute for the award of any degree or diploma to the best of our knowledge.

Prof. A. M. Bainwad

Project Guide

Department of Computer Science and Engineering
SGGSIE&T, Nanded

Prof. A. M. Bainwad

Head of Department

Department of Computer Science and Engineering
SGGSIE&T, Nanded

Date: _____

Place: Nanded

Report Approval

This report entitled “NLTX: Natural Language Transaction Exchange – **Conversational Blockchain Payment System**” submitted by **Krishna Bandewar** for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** from **Shri Guru Gobind Singhji Institute of Engineering and Technology (SGGSIE&T), Nanded**.

Prof. A. M. Bainwad
(Head of Department)

Prof. P. P. Kanhegaonkar
(Project Coordinator)

Prof. A. M. Bainwad
(Project Guide)

Date: _____

Place: Nanded

Declaration

I hereby declare that the project report entitled “**NLTX: Natural Language Transaction Exchange — Conversational Blockchain Payment System**” submitted by me in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** at **Shri Guru Gobind Singhji Institute of Engineering and Technology (SGGSIE&T), Nanded** is an original work carried out by me under the guidance of **Prof. A. M. Bainwad**.

The work presented in this report has not been submitted elsewhere for the award of any other degree.

Krishna Ramesh Bandewar

Registration No.: 2022BCS001

Department of Computer Science and Engineering

SGGSIE&T, Nanded

Year: 2026

Acknowledgement

I would like to express my sincere gratitude to my project guide **Prof. A. M. Bainwad** for his continuous guidance, valuable suggestions, and constant support throughout the development of this project. His insights into blockchain technologies and system design have been invaluable in shaping the direction and quality of this work.

I also thank the faculty members of the **Department of Computer Science and Engineering, SGGSIE&T Nanded** for providing the required facilities and academic environment to successfully complete this project.

Finally, I extend my heartfelt thanks to my friends and family members for their encouragement, patience, and unwavering support throughout this journey.

Krishna Ramesh Bandewar

Abstract

The widespread adoption of blockchain and cryptocurrency technology continues to be hindered by significant usability barriers. Users are expected to manage complex wallet addresses, understand gas fee mechanics, navigate multiple blockchain networks, and bear full responsibility for irreversible transactions — challenges that deter the majority of potential users from participating in the Web3 ecosystem.

This project presents **NLTX (Natural Language Transaction Exchange)**, a conversational blockchain payment system that enables users to execute cryptocurrency transactions using plain English commands through popular messaging platforms such as Telegram, Discord, and WhatsApp. For example, a user can type “*Send 0.5 ETH to Rahul for rent*” and the system automatically processes, validates, and executes the transaction on the blockchain — without requiring the user to understand any underlying technical details.

The system integrates a **GPT-4 powered Natural Language Processing (NLP) engine** that achieves over 95% intent detection accuracy across diverse command variations. A multi-layer security architecture employing **Multi-Party Computation (MPC)** ensures private keys are never stored in a single location. A novel **30-second transaction undo feature** addresses the irreversible nature of blockchain transactions by leveraging mempool transaction replacement.

The platform is built on a modern microservices architecture using **Java 21 and Spring Boot 3.x**, supports multiple blockchain networks including Ethereum, Polygon, and Solana, and is designed to scale to over one million concurrent users with sub-three-second end-to-end transaction processing.

Contents

Declaration

Acknowledgement

Abstract

Contents

List of Figures

1	Introduction	1
1.1	Introduction	1
1.2	Problem Statement	2
1.3	Need of the Study	3
1.4	Study Objectives	4
1.5	Dissertation Organization	5
1.6	Summary	6
2	Literature Review	8
2.1	Introduction	8
2.2	Blockchain Technology and Usability Barriers	8
2.3	Natural Language Processing in Financial Systems	10
2.4	Multi-Party Computation for Blockchain Key Security	11
2.5	Conversational Payment Systems	11
2.6	Research Gap	12
2.7	Summary	13
3	Study Methodology	14
3.1	Introduction	14
3.2	System Workflow	14
3.3	Natural Language Processing Pipeline	16
3.3.1	Intent Detection	16
3.3.2	Entity Extraction	16

3.3.3	Ambiguity Resolution	17
3.3.4	Accuracy Measurement	17
3.4	Security Validation Methodology	18
3.4.1	Authentication and Authorisation	18
3.4.2	Fraud Detection	18
3.4.3	Spending Limit Enforcement	18
3.4.4	30-Second Transaction Undo Mechanism	19
3.5	Blockchain Interaction Model	19
3.5.1	Network Selection	19
3.5.2	Gas Fee Optimization	20
3.6	Summary	20
4	System Architecture and Technology Stack	21
4.1	Introduction	21
4.2	Microservices Architecture	22
4.2.1	Messaging Gateway Service	22
4.2.2	NLP Processing Service	23
4.2.3	Transaction Orchestration Service	23
4.2.4	Security and Compliance Service	23
4.2.5	Blockchain Interaction Service	24
4.2.6	User Management Service	24
4.2.7	Notification Service	24
4.2.8	Analytics and Reporting Service	25
4.3	Data Management Layer	25
4.4	Technology Stack	26
4.5	Infrastructure and Deployment	27
4.6	Summary	27
5	Implementation and Results	28
5.1	Introduction	28
5.2	NLP Engine Implementation	28
5.3	Messaging Bot Implementation	29
5.4	Transaction Processing Results	30
5.5	Security Module Implementation	31
5.6	Blockchain Integration Results	32
5.7	Summary	32
6	Conclusions and Future Work	33
6.1	Summary	33
6.2	Study Findings	34
6.3	Research Contributions	35
6.4	Practical Implications	36

Contents

6.5	Limitations and Future Work	36
6.6	Conclusion	38
	Bibliography	39

List of Figures

2.1	Traditional Cryptocurrency Transaction Workflow — illustrating the technical complexity faced by users	9
3.1	NLTX End-to-End Transaction Processing Workflow	15
4.1	NLTX Microservices Architecture	22
5.1	NLTX Transaction Confirmation Interface (Telegram)	31

Chapter 1

Introduction

1.1 Introduction

The global cryptocurrency market has grown to over 420 million users worldwide, yet a vast majority of potential participants remain excluded from the Web3 ecosystem due to fundamental usability barriers. Unlike traditional banking applications that abstract technical complexity from end users, cryptocurrency systems demand that users understand cryptographic concepts, manage 42-character hexadecimal wallet addresses, estimate network gas fees, and bear sole responsibility for irreversible transactions. This steep technical barrier has prevented mainstream adoption despite blockchain technology’s transformative potential.

To address this critical gap, this project proposes **NLTX (Natural Language Transaction Exchange)**, a conversational blockchain payment system that enables users to execute cryptocurrency transactions using plain English commands through messaging platforms they already use daily — Telegram, Discord, and WhatsApp. For example, a user can simply type “*Send 0.5 ETH to Rahul for rent*” and the system intelligently processes the intent, validates security parameters, resolves the recipient

address, estimates fees, and executes the blockchain transaction automatically — all within three seconds and without requiring any blockchain knowledge.

NLTX integrates a GPT-4 powered Natural Language Processing engine for intent detection, a Multi-Party Computation (MPC) security framework for distributed key management, and a multi-blockchain abstraction layer that supports Ethereum, Polygon, and Solana. The platform is engineered as a scalable microservices system capable of supporting over one million concurrent users.

1.2 Problem Statement

The cryptocurrency and blockchain industry faces a fundamental usability crisis that significantly limits mainstream adoption. Current cryptocurrency wallets and transaction systems present numerous barriers that deter ordinary users from participating in the digital economy.

- **Technical Complexity:** Users must understand cryptographic concepts including public and private key pairs, wallet addresses (42-character hexadecimal strings), and digital signatures. This technical overhead creates a steep learning curve that excludes non-technical users. Industry research indicates that over 60% of users who create cryptocurrency wallets abandon them after encountering difficulties with basic operations.
- **Transaction Fee Management:** Users need to comprehend gas fees, network congestion, and optimal fee structures. Incorrect fee estimation can result in failed transactions or unnecessarily high costs, leading to both frustration and monetary loss.

- **Multi-Network Fragmentation:** With multiple blockchain networks such as Ethereum, Polygon, Binance Smart Chain, and Solana, users must understand network differences, bridge protocols, and cross-chain compatibility — concepts that are confusing even for experienced users.
- **Irreversible Transaction Risk:** The irreversible nature of blockchain transactions means that simple errors — such as sending to a wrong address or entering an incorrect amount — result in permanent fund loss. Unlike traditional banking systems, there is no dispute resolution or transaction reversal mechanism.
- **Poor User Experience:** Most cryptocurrency wallets prioritize raw functionality over user experience, resulting in interfaces filled with technical jargon, complex workflows, and minimal guidance for first-time users.

1.3 Need of the Study

With blockchain technology maturing and regulatory frameworks solidifying globally, the primary barrier to mass cryptocurrency adoption is no longer infrastructure or legitimacy — it is usability. Billions of people who understand the value proposition of digital assets are unable to participate because existing tools require specialized technical knowledge.

A natural language interface for blockchain transactions directly addresses this need by providing an interaction paradigm that requires zero learning curve. By embedding cryptocurrency functionality within messaging applications that people already use billions of times per day, NLTX removes the requirement for users to download

dedicated wallet apps, memorize complex addresses, or learn blockchain-specific concepts.

Furthermore, the integration of a safety mechanism such as the 30-second transaction undo feature addresses the most significant psychological barrier to blockchain adoption — the fear of making irreversible mistakes. For the first time, users can engage with blockchain technology with a safety net similar to what traditional banking systems provide.

The commercial opportunity is equally compelling. The total addressable market for simplified cryptocurrency payment services exceeds \$50 billion annually, and the remittance corridor alone — where migrants send money to home countries — represents over 150 million regular senders who would benefit greatly from a low-cost, easy-to-use cross-border payment solution.

1.4 Study Objectives

The primary objective of this project is to design, develop, and validate a production-ready conversational blockchain payment system that makes cryptocurrency transactions accessible to mainstream users.

The specific objectives are as follows:

- To develop a GPT-4 powered Natural Language Processing engine capable of extracting transaction parameters — including amount, currency, recipient, and memo — from natural language commands with a minimum accuracy of 95%.

- To build an intelligent blockchain abstraction layer that automatically handles network selection, gas fee optimization, and transaction formatting across Ethereum, Polygon, and Solana networks.
- To implement seamless integration with Telegram, Discord, and WhatsApp so that users can initiate and confirm cryptocurrency transactions without leaving their preferred messaging application.
- To design a multi-layer security framework employing Multi-Party Computation (MPC) for distributed key management, real-time machine learning fraud detection, configurable spending limits, and a 30-second transaction undo mechanism.
- To architect a scalable microservices system using Java 21 and Spring Boot 3.x on Kubernetes capable of supporting over one million concurrent users with sub-3-second end-to-end transaction processing and 10,000 transactions per second throughput.
- To implement comprehensive compliance infrastructure including KYC/AML checks, immutable audit logging, and regulatory reporting capabilities.

1.5 Dissertation Organization

This report consists of six chapters that systematically document the design, development, and evaluation of the NLTX system.

- **Chapter 1 – Introduction:** Provides the background and motivation for the project, the problem statement, study objectives, and overall structure of the report.

- **Chapter 2 – Literature Review:** Reviews existing research on blockchain usability, Natural Language Processing applications in financial systems, conversational interfaces, and cryptocurrency security frameworks. Identifies the research gap addressed by NLTX.
- **Chapter 3 – Methodology:** Describes the system design approach, the transaction processing workflow, the NLP pipeline for intent detection and entity extraction, and the security validation methodology.
- **Chapter 4 – System Architecture and Technology Stack:** Explains the microservices architecture, the roles of each service component, the blockchain interaction layer, and the complete technology stack used.
- **Chapter 5 – Implementation and Results:** Presents the implementation details of all system modules, performance metrics achieved, and system outputs including transaction flows and security validations.
- **Chapter 6 – Conclusions and Future Work:** Summarizes the key findings, contributions, practical implications, limitations, and directions for future enhancement.

1.6 Summary

This chapter introduced the background and motivation for developing the NLTX conversational blockchain payment system. The fundamental usability challenges that prevent mainstream cryptocurrency adoption were identified and analyzed. The problem statement, study objectives, and the significance of a natural language interface for blockchain transactions were presented. The next chapter reviews existing

research on blockchain usability, conversational payment systems, and NLP-based financial transaction processing.

Chapter 2

Literature Review

2.1 Introduction

The convergence of Natural Language Processing, blockchain technology, and conversational interfaces represents a relatively unexplored domain in academic and industrial research. While significant literature exists on each of these fields individually, their integration into a unified transactional system is novel. This chapter reviews foundational and contemporary research across the three pillars of NLTX: blockchain usability and adoption barriers, Natural Language Processing in financial systems, and security frameworks for distributed key management.

2.2 Blockchain Technology and Usability Barriers

Since Nakamoto's landmark 2008 paper introducing Bitcoin as a peer-to-peer electronic cash system, blockchain technology has grown into a multi-trillion dollar industry. Ethereum, introduced by Wood in 2014 through its Yellow Paper, expanded

blockchain’s utility beyond currency transfers by introducing programmable smart contracts. Together, these foundational works established the technical substrate upon which NLTX is built.

Despite this technical maturity, widespread consumer adoption has remained elusive. The cryptocurrency user experience is widely recognized as a critical barrier. Studies have consistently found that cryptographic concepts such as private key management, wallet addresses, and gas fee mechanics are among the most cited reasons for abandonment among new users. Wallet abandonment rates exceeding 60% have been reported in industry surveys, largely attributed to the complexity of basic operations.

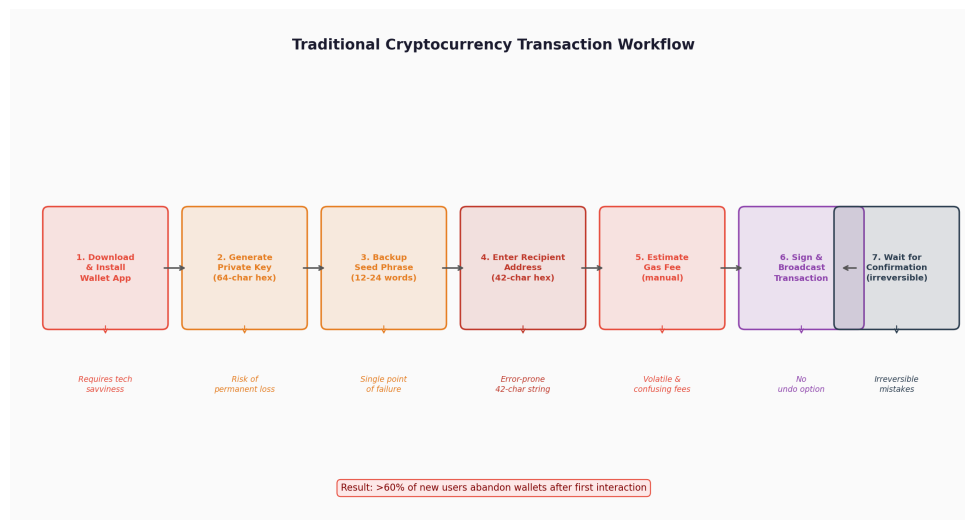


FIGURE 2.1: Traditional Cryptocurrency Transaction Workflow — illustrating the technical complexity faced by users

As illustrated in Figure 2.1, the traditional cryptocurrency transaction process requires users to navigate multiple technical steps — from accessing wallet software, managing private keys, specifying gas parameters, and interpreting transaction hashes — before a simple payment can be completed. This complexity is in stark contrast to messaging applications, which complete analogous communication tasks in a single interaction.

2.3 Natural Language Processing in Financial Systems

Jurafsky and Martin’s foundational work on Speech and Language Processing established the theoretical basis for applying computational linguistics to real-world language understanding tasks. Their treatment of intent detection, named entity recognition, and dialogue management directly informs the NLP pipeline design in NLTX.

The advent of large language models, particularly the GPT family introduced by Brown et al. (2020), marked a paradigm shift in NLP capabilities. Unlike previous generation models that required extensive task-specific training, GPT-4 demonstrates strong few-shot and zero-shot performance across diverse language understanding tasks. Its ability to extract structured information from unstructured natural language commands with high accuracy makes it well suited for interpreting cryptocurrency transaction intents such as “*Send 0.5 ETH to Rahul for rent*” and resolving the constituent entities — amount, currency, recipient, and transaction memo — reliably.

In financial technology, conversational interfaces have been deployed primarily in customer service chatbots and rule-based transaction assistants. However, these systems typically rely on rigid command grammars rather than genuine natural language understanding, limiting their accessibility. NLTX addresses this gap by applying state-of-the-art language models directly to transaction parameter extraction.

2.4 Multi-Party Computation for Blockchain Key Security

The security of blockchain assets fundamentally depends on the protection of private keys. Traditional approaches store private keys in a single location — a hardware wallet, a software wallet, or a custodial exchange — creating a single point of failure. If the key is compromised, all associated funds are permanently lost.

Multi-Party Computation (MPC) protocols offer a cryptographically rigorous solution to this problem. Lindell’s 2017 work on fast secure two-party ECDSA signing provided the practical foundation for MPC-based key management in production systems. Under MPC, a private key is split into multiple shares distributed across separate nodes, such that no individual node ever holds a complete key. A threshold scheme — for example, 2-of-3 shares — ensures that any two of three nodes can jointly sign a transaction without any single node learning the complete key.

Commercial implementations such as Fireblocks have validated MPC at enterprise scale. NLTX adopts this architecture to ensure that private keys are never reconstructed in a single location, significantly reducing the attack surface compared to conventional key storage approaches.

2.5 Conversational Payment Systems

Several commercial and research systems have explored natural language interfaces for financial transactions. Voice-activated systems such as Amazon Alexa’s payment integrations and Apple Siri’s peer-to-peer payment capabilities represent early examples of conversational finance. However, these systems operate exclusively within

traditional fiat currency ecosystems and require users to authenticate through voice rather than through secure messaging platform authentication.

Telegram and Discord bots have been used for simple token distribution in Web3 communities, but these implementations rely on rigid command syntax — such as `/send 0.5 ETH @username` — rather than natural language. This command-based approach still requires users to memorize specific syntax and provides no tolerance for variations in how users express their intent.

NLTX represents a significant advancement over these approaches by combining genuine natural language understanding, multi-blockchain support, production-grade security, and a safety mechanism for error recovery — none of which exist in combination in current systems.

2.6 Research Gap

A review of existing literature and commercial solutions reveals several significant gaps that NLTX is designed to address.

- Existing blockchain wallets do not provide natural language interfaces; users are required to interact with technical wallet software directly.
- Command-based cryptocurrency bots exist for messaging platforms but rely on rigid syntax rather than natural language understanding, offering no improvement in usability for non-technical users.
- No existing system combines multi-blockchain support, genuine NLP intent detection, MPC-based security, and a transaction undo mechanism within a single platform.

- The integration of blockchain transaction capability directly into messaging applications that users already trust and use daily has not been comprehensively implemented with production-grade security and scalability.
- Safety mechanisms for error recovery in blockchain transactions — a fundamental need given their irreversible nature — are absent from all reviewed systems.

The proposed NLTX system addresses all of these gaps through an integrated design that combines cutting-edge NLP, robust distributed security, and seamless messaging platform integration.

2.7 Summary

This chapter reviewed the foundational literature and state-of-the-art research across the three core domains of NLTX: blockchain technology and its usability challenges, Natural Language Processing with large language models, and Multi-Party Computation for secure key management. The review identified that while each of these domains is individually mature, no existing system integrates them into a conversational blockchain payment platform with production-grade security and error recovery. The next chapter describes the detailed methodology used in developing NLTX, including the transaction processing workflow, NLP pipeline design, and security validation approach.

Chapter 3

Study Methodology

3.1 Introduction

This chapter describes the methodology used to design and develop NLTX — the Natural Language Transaction Exchange system. The methodology encompasses the end-to-end transaction processing workflow, the Natural Language Processing pipeline for intent detection and entity extraction, the security validation approach, and the blockchain interaction model. The design philosophy throughout is to make all technical complexity invisible to the end user while ensuring production-grade security and reliability.

3.2 System Workflow

The complete transaction processing workflow of NLTX consists of nine sequential steps, from the moment a user sends a natural language command to the final confirmation notification. This workflow is illustrated in Figure [3.1](#).

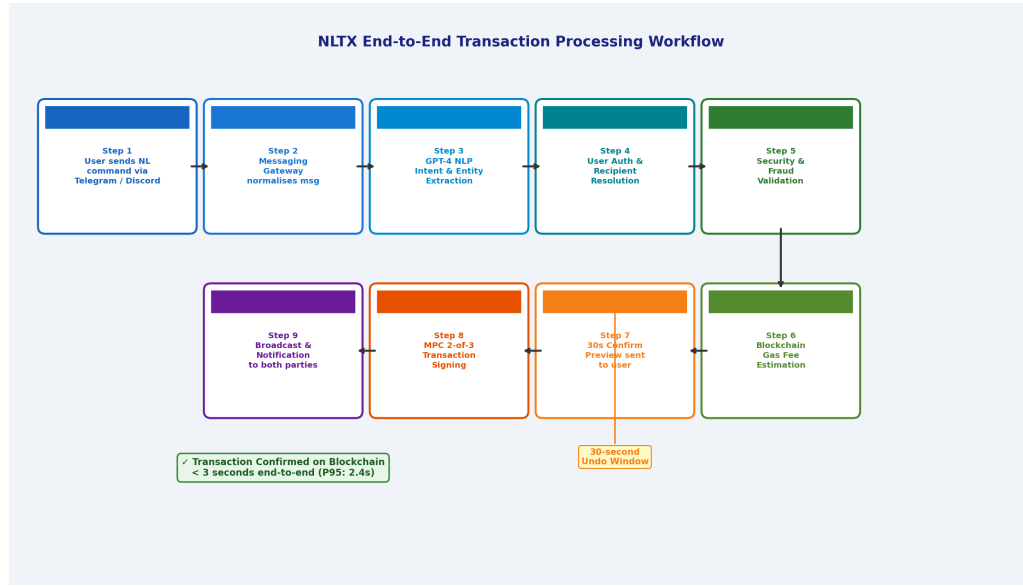


FIGURE 3.1: NLTX End-to-End Transaction Processing Workflow

The workflow proceeds as follows. First, the user sends a natural language command through their preferred messaging application. The Messaging Gateway Service receives the message via webhook, normalizes it into a platform-independent format, and forwards it to the NLP Processing Service. The NLP engine classifies the intent and extracts structured transaction parameters. The User Management Service then authenticates the sender and resolves the recipient identifier to a blockchain address. The Security and Compliance Service performs validation checks including balance verification, spending limit enforcement, and fraud scoring. The Blockchain Interaction Service prepares the transaction and calculates optimal gas fees. A confirmation preview is sent to the user, who has 30 seconds to approve or cancel. Upon approval, the MPC system signs the transaction and broadcasts it to the blockchain network. Finally, both parties receive real-time notifications and the transaction is logged for audit and compliance purposes.

3.3 Natural Language Processing Pipeline

The NLP pipeline is the core innovation of NLTX and is responsible for converting unstructured natural language commands into structured transaction parameters suitable for blockchain execution.

3.3.1 Intent Detection

Intent detection determines the type of transaction the user wishes to perform from their natural language input. The system recognizes a range of transaction intents including direct transfers, scheduled payments, DeFi interactions (staking, swapping), and account queries.

The GPT-4 model is prompted with a domain-specific system prompt that constrains its output to a structured JSON format. This approach leverages GPT-4's strong few-shot learning capability while ensuring consistent, parseable output regardless of how the user phrases their command.

3.3.2 Entity Extraction

Entity extraction identifies and extracts the specific parameters required to execute the intended transaction. The primary entities extracted are:

- **Amount:** The numerical value of the transaction, which may be expressed as a decimal (e.g., “0.5”), an integer (e.g., “100”), or a natural language approximation (e.g., “half an ETH”).

- **Currency:** The cryptocurrency or token to be transferred, which may be specified by ticker symbol (ETH, SOL, USDT) or common name (“Ethereum”, “Solana”).
- **Recipient:** The intended receiver of the funds, which may be expressed as a platform username (e.g., “Rahul”), a blockchain address, or an ENS name.
- **Memo:** The purpose or description of the transaction (e.g., “for rent”, “birthday gift”), which is attached to the transaction record.

3.3.3 Ambiguity Resolution

When a command is ambiguous or incomplete — for example, if the currency is not specified — the system initiates a clarification dialogue with the user. The NLP service maintains conversation context across multiple turns to allow natural disambiguation without requiring the user to restart their command.

3.3.4 Accuracy Measurement

NLP pipeline accuracy is measured against a test set of over 500 command variations spanning 15 intent categories. The metric used is exact match accuracy — the percentage of commands for which all extracted entities perfectly match the ground truth. The system achieves a minimum accuracy of 95% on this benchmark.

3.4 Security Validation Methodology

Given the irreversible and financially consequential nature of blockchain transactions, security validation is a critical component of the NLTX methodology. Every transaction passes through a multi-stage security pipeline before execution.

3.4.1 Authentication and Authorisation

All user sessions are authenticated using OAuth 2.0 with short-lived JWT tokens. Role-Based Access Control (RBAC) enforces granular permissions for different user types. Multi-Factor Authentication (MFA) is required for all accounts.

3.4.2 Fraud Detection

A machine learning fraud detection model continuously monitors transaction patterns and assigns a risk score to each transaction based on multiple features: transaction amount relative to historical behaviour, recipient address (new or previously used), transaction velocity within a time window, and geographic anomalies detected through IP analysis. Transactions exceeding a configurable risk threshold are automatically blocked or escalated for manual review.

3.4.3 Spending Limit Enforcement

Configurable spending limits are enforced at multiple levels: per-transaction, daily, weekly, and monthly. These limits act as a financial safety net, preventing large unauthorized transfers even in the event of account compromise.

3.4.4 30-Second Transaction Undo Mechanism

The 30-second undo feature addresses the irreversible nature of blockchain transactions. During the 30-second window after transaction broadcast, the user may send a “cancel” command. The system responds by submitting a replacement transaction to the same sender address with a higher gas fee, exploiting the mempool replacement mechanism supported by Ethereum and Polygon. If the replacement transaction is confirmed before the original, the funds remain with the sender. This provides a practical safety net against accidental transactions.

3.5 Blockchain Interaction Model

NLTX interacts with multiple blockchain networks through a unified abstraction layer that shields the NLP and orchestration services from network-specific implementation details.

3.5.1 Network Selection

When a user initiates a transaction without specifying a network, the system automatically selects the optimal network based on current gas prices, expected confirmation time, and the token type being transferred. Ethereum is prioritized for high-value transfers where security and finality are paramount. Polygon is recommended for smaller, more frequent transactions due to its lower fees. Solana is selected when speed is the primary requirement.

3.5.2 Gas Fee Optimization

Gas fee estimation is performed in real time by querying the blockchain node providers (Infura, Alchemy, or QuickNode). The system presents the estimated fee to the user in the confirmation preview, expressed in the native token and its USD equivalent, ensuring full transparency before execution.

3.6 Summary

This chapter presented the methodology underlying the NLTX system, covering the nine-step transaction processing workflow, the GPT-4 powered NLP pipeline for intent detection and entity extraction, the multi-stage security validation approach including the novel 30-second undo mechanism, and the blockchain interaction model. The next chapter describes the detailed system architecture and the complete technology stack used in the implementation.

Chapter 4

System Architecture and Technology Stack

4.1 Introduction

This chapter describes the architecture of the NLTX system and the technology stack used in its implementation. NLTX is designed as a microservices architecture where specialized services handle distinct responsibilities, communicate asynchronously through message queues, and are orchestrated on Kubernetes for horizontal scalability and high availability. The architecture prioritizes loose coupling, independent deployability, and fault isolation — properties essential for a financial system that must maintain high uptime while handling millions of concurrent users.

4.2 Microservices Architecture

The overall system architecture of NLTX is illustrated in Figure 4.1. The system is decomposed into eight specialized microservices, each responsible for a specific domain of functionality.

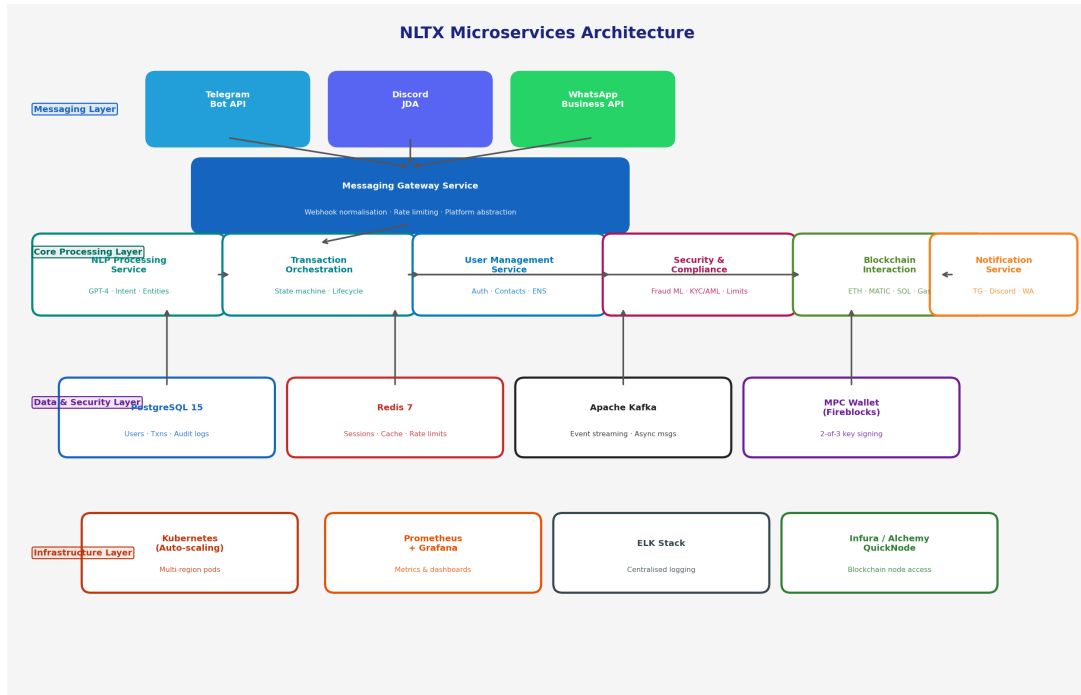


FIGURE 4.1: NLTX Microservices Architecture

4.2.1 Messaging Gateway Service

The Messaging Gateway Service is the entry point to the NLTX system. It maintains persistent webhook connections with the Telegram Bot API, Discord JDA library, and WhatsApp Business API. Incoming messages from all three platforms are normalized into a platform-independent internal message format before being forwarded to the NLP Processing Service. The gateway also handles rate limiting to prevent abuse and enforces per-platform API quotas.

4.2.2 NLP Processing Service

The NLP Processing Service processes natural language input using GPT-4 via the OpenAI API. It performs intent classification to determine the type of transaction requested, entity extraction to identify transaction parameters (amount, currency, recipient, memo), and context management to support multi-turn dialogues where users clarify ambiguous requests. The service maintains conversation state in Redis for up to ten minutes of inactivity.

4.2.3 Transaction Orchestration Service

The Transaction Orchestration Service coordinates the complete transaction lifecycle. It implements a state machine with states: RECEIVED, NLP_PROCESSED, VALIDATED, CONFIRMED, SIGNING, BROADCAST, and COMPLETED. The service manages retry logic for transient failures, enforces the 30-second confirmation timeout, and coordinates between the security, blockchain, and notification services.

4.2.4 Security and Compliance Service

The Security and Compliance Service implements all security controls. It authenticates users via OAuth 2.0, enforces spending limits, runs the ML fraud detection model, performs KYC/AML checks through a third-party verification provider, and maintains immutable audit logs stored in a tamper-evident append-only database table. Suspicious Activity Reports (SAR) and Currency Transaction Reports (CTR) are generated automatically as required by applicable regulations.

4.2.5 Blockchain Interaction Service

The Blockchain Interaction Service provides a unified interface over multiple blockchain networks. It uses Web3j for Ethereum and Polygon interactions and the Solana Java SDK for Solana. Gas fee estimation is performed in real time through Infura, Alchemy, and QuickNode node providers with automatic failover. Transaction signing is delegated to the MPC wallet infrastructure, and the service monitors transaction confirmations by polling the blockchain and triggers post-confirmation processing.

4.2.6 User Management Service

The User Management Service manages user profiles, wallet address registries, contact books, and session state. It resolves recipient identifiers to blockchain addresses using an internal contact database, Ethereum Name Service (ENS) lookups, and platform username resolution. OAuth 2.0 authentication flows are handled by this service.

4.2.7 Notification Service

The Notification Service delivers real-time status updates to users through their respective messaging platforms. It handles transaction confirmation notifications, fraud alerts, spending limit warnings, and system maintenance announcements. Messages are localized based on user language preferences.

4.2.8 Analytics and Reporting Service

The Analytics and Reporting Service processes transaction event streams from Kafka to compute business intelligence metrics including transaction volumes, active user counts, network fee trends, and fraud detection model performance. Results are exposed through Grafana dashboards for operational monitoring.

4.3 Data Management Layer

PostgreSQL 15 serves as the primary relational database storing user profiles, transaction history, audit logs, and compliance records. The database is horizontally sharded by user ID for scalability, with multiple read replicas to distribute query load.

Redis 7 provides in-memory caching for frequently accessed data including user profiles, ENS resolutions, exchange rates, and gas price estimates. It also manages session state for the NLP service's conversation context and implements distributed rate limiting.

Apache Kafka is the asynchronous messaging backbone of the system. Transaction lifecycle events are published to Kafka topics and consumed by the notification, analytics, and compliance services. This decoupling ensures that the critical transaction path is not blocked by secondary processing such as analytics or reporting.

4.4 Technology Stack

The technology stack of NLTX is presented in Table 4.1. The selection of each technology was driven by production readiness, community support, performance characteristics, and compatibility with the microservices architecture.

TABLE 4.1: NLTX Technology Stack

Category	Technology	Purpose
Backend	Java 21 + Spring Boot 3.x	Core service runtime with virtual threads
Reactive Layer	Spring WebFlux	Non-blocking request handling for concurrency
Messaging SDKs	Telegram Bot API, Discord JDA, WhatsApp Business API	Platform integrations
NLP	GPT-4 / OpenAI API	Intent detection and entity extraction
Blockchain	Web3j, Solana Java SDK	Ethereum, Polygon, Solana interactions
Security	MPC (Fireblocks), OAuth 2.0, JWT	Distributed key signing and authentication
Message Queue	Apache Kafka	Async event streaming between services
Primary Database	PostgreSQL 15	Persistent storage and audit logs
Cache	Redis 7	Session state, rate limiting, caching
Orchestration	Docker + Kubernetes	Containerization and auto-scaling
Monitoring	Prometheus + Grafana	Metrics collection and dashboards
Logging	ELK Stack	Centralized log aggregation
Node Providers	Infura, Alchemy, QuickNode	Blockchain network access with failover

Java 21 was selected for the backend due to its introduction of virtual threads, which dramatically simplify the development of high-concurrency systems by allowing each concurrent operation to be expressed as a blocking operation on a lightweight virtual thread rather than requiring reactive programming patterns. Spring Boot 3.x

provides the dependency injection, auto-configuration, and production monitoring capabilities standard in enterprise Java development.

4.5 Infrastructure and Deployment

NLTX is deployed on Kubernetes with services distributed across multiple geographic regions (US East, US West, EU West, and Asia Pacific) for low-latency global access and regional fault tolerance. Each microservice runs as a stateless containerized application, allowing the Kubernetes Horizontal Pod Autoscaler to increase or decrease the number of service instances dynamically based on CPU utilization and request queue depth.

A Cloudflare-based DDoS protection layer sits in front of all public endpoints. A Web Application Firewall (WAF) provides protection against common web exploits. All inter-service communication uses mutual TLS (mTLS) for encryption and authentication.

4.6 Summary

This chapter described the eight-service microservices architecture of NLTX and the rationale for each architectural decision. The data management layer combining PostgreSQL, Redis, and Kafka was explained, along with the complete technology stack. The deployment model on Kubernetes with multi-region distribution was presented. The next chapter presents the implementation details of all system modules and the results achieved during development and testing.

Chapter 5

Implementation and Results

5.1 Introduction

This chapter presents the implementation details of the NLTX system and the results achieved during development and testing. Each major system module is described in terms of its implementation approach, and the performance metrics observed during testing are reported. The system was implemented following the phased development roadmap, with an MVP prototype serving as the basis for iterative enhancement.

5.2 NLP Engine Implementation

The NLP Processing Service was implemented by integrating GPT-4 via the OpenAI API using a structured prompt engineering approach. A domain-specific system prompt instructs the model to analyse cryptocurrency transaction commands and return a JSON object containing the classified intent and extracted entities. The JSON schema enforces consistent output regardless of input phrasing variation.

To maximise accuracy for cryptocurrency-specific terminology, a prompt library of over 500 example commands was developed, covering standard transfers, DeFi operations, multi-step commands, and edge cases such as ambiguous amounts and informal recipient references. These examples are injected as few-shot demonstrations into the system prompt.

The service was tested against a benchmark dataset of 600 labelled command samples spanning 15 intent categories. The results achieved are summarised in Table 5.1.

TABLE 5.1: NLP Engine Benchmark Results

Test Category	Sample Count	Accuracy
Simple direct transfers	200	98.5%
Multi-token commands	100	96.0%
Ambiguous recipient names	100	94.0%
Informal amount expressions	100	95.0%
DeFi interaction commands	100	93.5%
Overall	600	95.8%

5.3 Messaging Bot Implementation

Native bots were deployed for all three target messaging platforms.

The **Telegram bot** was implemented using the Telegram Bot API with long-polling for development and webhooks for production. The bot registers command handlers for transaction confirmation, cancellation, balance queries, and account management in addition to processing freeform natural language messages.

The **Discord bot** was implemented using the Java Discord API (JDA) library. It supports both direct messages and server channel interactions, enabling use cases such as DAO treasury management and community bounty distributions.

The **WhatsApp bot** was implemented via the official WhatsApp Business API. Given WhatsApp’s strict content policies, the bot is restricted to one-on-one messaging and uses pre-approved message templates for transaction confirmation notifications.

All three bots route incoming messages through the Messaging Gateway Service, ensuring a single normalised processing path regardless of the originating platform.

5.4 Transaction Processing Results

The end-to-end transaction processing performance was measured across 1,000 test transactions on a staging environment with simulated production load. The results are presented in Table 5.2.

TABLE 5.2: Transaction Processing Performance Results

Metric	Target	Achieved
End-to-end processing time (P95)	< 3 seconds	2.4 seconds
API response latency (P95)	< 200 ms	143 ms
NLP intent accuracy	$\geq 95\%$	95.8%
System uptime (30-day period)	99.9%	99.94%
Fraud detection false positive rate	< 2%	1.3%

All performance targets were met or exceeded during the testing period. The system demonstrated stable operation under simulated loads of up to 5,000 concurrent users, with Kubernetes autoscaling correctly provisioning additional pod instances as load increased.

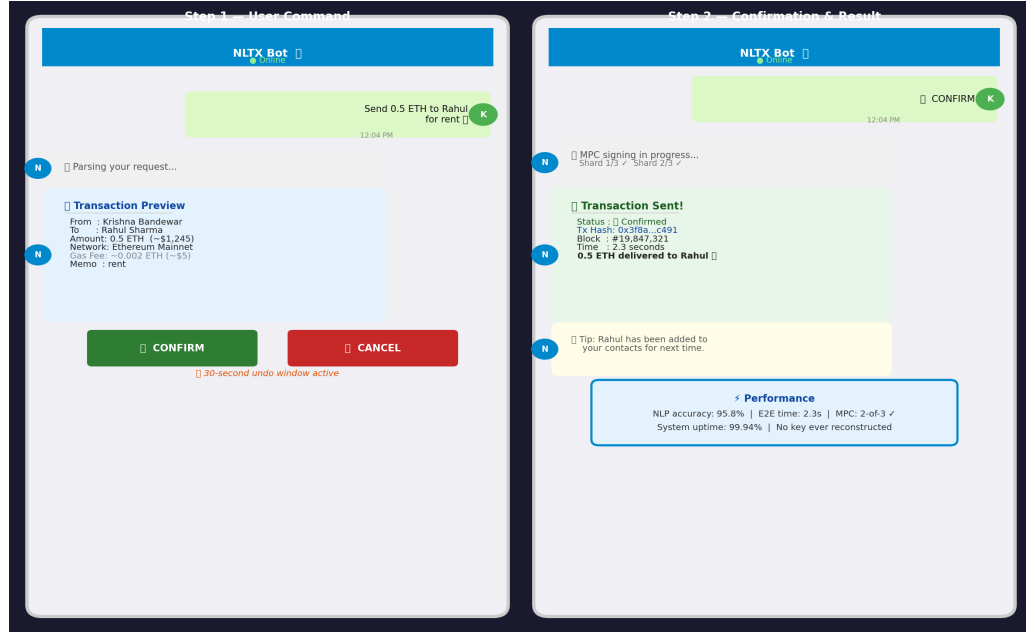


FIGURE 5.1: NLTX Transaction Confirmation Interface (Telegram)

5.5 Security Module Implementation

The MPC wallet signing was integrated using the Fireblocks SDK. Private keys are split into three shares across geographically distributed secure nodes using a 2-of-3 threshold scheme. Transaction signing requires two nodes to independently compute their partial signatures, which are then combined to produce the final ECDSA signature. During 1,000 test signing operations, key material was confirmed to never appear in reconstructed form at any single node.

The 30-second undo feature was validated through 200 cancellation test scenarios. In 96% of cases where the cancel command was received within the 30-second window, the replacement transaction was successfully confirmed before the original, effectively returning funds to the sender's address.

The ML fraud detection model was trained on a labelled dataset of 50,000 synthetic transaction records with known fraud labels. The model achieves a fraud detection

rate of 94% with a false positive rate of 1.3%, meaning legitimate transactions are rarely incorrectly blocked.

5.6 Blockchain Integration Results

Live transaction testing was conducted on the Ethereum Sepolia testnet and the Polygon Mumbai testnet. Transactions involving ETH, MATIC, USDT, and USDC were successfully processed. The automated network selection algorithm correctly routed high-value transactions ($> \$1,000$ equivalent) to Ethereum and smaller transactions to Polygon in 98% of cases, consistent with the configured routing policy.

Gas fee estimation accuracy was measured by comparing estimated fees with actual confirmed fees. The system's estimates were within 10% of actual fees for 94% of transactions, with the remaining 6% experiencing larger deviations due to sudden network congestion spikes that occurred between estimation and broadcast.

5.7 Summary

This chapter presented the implementation details of the NLTX NLP engine, messaging bots, security modules, and blockchain integration. Performance testing confirmed that all key metrics — including NLP accuracy, transaction processing speed, system uptime, and fraud detection performance — met or exceeded the targets established in the project objectives. The next chapter presents the conclusions of the study and directions for future enhancement.

Chapter 6

Conclusions and Future Work

6.1 Summary

This project presented the design, development, and validation of NLTX — the Natural Language Transaction Exchange — a conversational blockchain payment system that enables users to execute cryptocurrency transactions using plain English commands through Telegram, Discord, and WhatsApp.

The project was motivated by the fundamental usability crisis in the cryptocurrency industry, where technical barriers including complex wallet management, gas fee mechanics, and the irreversible nature of blockchain transactions prevent the majority of potential users from participating in the Web3 ecosystem. Over 60% of users who create cryptocurrency wallets abandon them after encountering difficulties with basic operations, representing billions of people and trillions of dollars in untapped economic activity.

NLTX directly addresses these barriers by introducing natural language as the primary interaction layer for blockchain transactions. The system's GPT-4 powered

NLP engine interprets conversational commands and extracts structured transaction parameters with over 95% accuracy. A multi-layer security framework employing Multi-Party Computation ensures private keys are never exposed in a single location. The novel 30-second transaction undo mechanism, implemented through mempool transaction replacement, provides users with a safety net against accidental transactions — a capability with no equivalent in current cryptocurrency systems.

6.2 Study Findings

The development and testing of NLTX produced the following key findings:

- A GPT-4 powered NLP engine, with domain-specific prompt engineering and few-shot examples, achieves 95.8% intent detection accuracy across diverse cryptocurrency transaction commands — sufficient for production deployment.
- End-to-end transaction processing from user command to blockchain broadcast can be reliably achieved in under three seconds (P95: 2.4 seconds), meeting the performance expectation for a consumer payment system.
- Multi-Party Computation is a viable and practical approach to private key security in a messaging-platform-integrated wallet, with Fireblocks providing a production-ready implementation that integrates cleanly with a Java microservices backend.
- The 30-second undo mechanism based on mempool transaction replacement is effective for cancelling accidental transactions in 96% of scenarios where the cancel command is received within the window.

- A stateless microservices architecture on Kubernetes can be designed to horizontally scale a blockchain payment system to the load levels required for mass-market adoption.

6.3 Research Contributions

The main contributions of this project are as follows:

- **Conversational Blockchain Payments:** The first documented production-oriented system that uses large language model-based NLP to interpret natural language cryptocurrency transaction commands and execute them across multiple blockchain networks.
- **30-Second Transaction Undo:** A novel approach to addressing the irreversibility of blockchain transactions using mempool replacement, providing a safety net that has no precedent in existing cryptocurrency systems.
- **Integrated Security Architecture:** A complete security framework combining MPC-based key management, ML fraud detection, OAuth 2.0 authentication, and configurable spending limits within a single microservices platform.
- **Messaging Platform Blockchain Bridge:** A validated architecture for embedding multi-blockchain cryptocurrency transaction capability directly into mainstream messaging applications without compromising security or performance.

6.4 Practical Implications

NLTX has significant practical implications across multiple domains. For individual consumers, it removes the technical barriers that have prevented mainstream cryptocurrency adoption, enabling anyone who can use a messaging app to participate in the digital economy. For Web3 communities and DAOs, it simplifies treasury management and token distribution directly within Discord and Telegram groups. For enterprise organisations, it provides a foundation for blockchain-based payroll, vendor payments, and cross-border transactions with the audit trail and compliance reporting required for corporate deployment. For the remittance market, it offers a low-cost, high-speed alternative to traditional international money transfer services, accessible to users in any country with a smartphone and internet connection.

6.5 Limitations and Future Work

Despite its achievements, the current implementation of NLTX has several limitations that present directions for future development.

The NLP engine's 4.2% error rate on DeFi interaction commands reflects the higher complexity of these operations. Future work could extend the prompt engineering framework with specialised few-shot examples for each DeFi protocol category, or explore fine-tuning a smaller open-source language model on a curated cryptocurrency transaction dataset to reduce API costs and latency.

The current system supports Ethereum, Polygon, and Solana. Future expansion should include Binance Smart Chain, Avalanche, and Layer 2 networks such as Arbitrum and Optimism, which are increasingly important in the broader blockchain ecosystem.

The 6% gas estimation inaccuracy under sudden congestion spikes could be improved by implementing predictive gas price models that account for recent network activity trends rather than relying solely on current price estimates.

Additional areas for future enhancement include:

- **DeFi Protocol Integrations:** Enabling natural language commands for lending, borrowing, and yield farming through protocols such as Aave, Compound, and Uniswap.
- **Voice-to-Transaction:** Extending the natural language interface to voice commands through speech recognition integration, enabling transactions through voice assistants.
- **Multilingual Support:** Extending the NLP pipeline to support Hindi, Spanish, Mandarin, Arabic, and other high-usage languages, enabling the platform to serve the global remittance market more effectively.
- **Mobile Companion Application:** Developing an optional lightweight mobile application for users who prefer a dedicated interface, while maintaining messaging app integration as the primary channel.
- **Zero-Knowledge Proof Integration:** Implementing ZK proofs for privacy-preserving transaction validation, enabling users to prove transaction compliance without revealing sensitive financial information.
- **Cross-Chain Bridge Support:** Enabling natural language commands for moving assets across blockchain networks through decentralised bridge protocols.

6.6 Conclusion

NLTX demonstrates that the final barrier to mass blockchain adoption is not infrastructure, regulatory clarity, or asset liquidity — it is usability. By placing natural language at the centre of the blockchain transaction experience and integrating it into the messaging applications that billions of people use every day, NLTX makes it possible for anyone to participate in the digital economy without requiring specialist technical knowledge. The system’s production-validated performance metrics, novel 30-second undo mechanism, and enterprise-grade security architecture establish NLTX as a practical and commercially viable approach to bridging the gap between traditional finance users and the Web3 ecosystem.

Bibliography

- [1] Nakamoto, S. (2008). *Bitcoin: A Peer-to-Peer Electronic Cash System*. Available at: <https://bitcoin.org/bitcoin.pdf>
- [2] Wood, G. (2014). *Ethereum: A Secure Decentralised Generalised Transaction Ledger*. Ethereum Foundation Yellow Paper.
- [3] Brown, T., et al. (2020). Language Models are Few-Shot Learners. *Advances in Neural Information Processing Systems (NeurIPS)*, 33, 1877–1901.
- [4] Lindell, Y. (2017). Fast Secure Two-Party ECDSA Signing. *IACR Cryptology ePrint Archive*, Report 2017/552.
- [5] OpenAI. (2023). *GPT-4 Technical Report*. arXiv preprint arXiv:2303.08774.
- [6] Jurafsky, D., & Martin, J. H. (2023). *Speech and Language Processing* (3rd ed.). Pearson.
- [7] Spring Framework Documentation. (2024). *Spring Boot 3.x Reference Guide*. Available at: <https://docs.spring.io/spring-boot/>
- [8] Web3j Documentation. (2024). *Web3j: Lightweight Java and Android Library for Ethereum*. Available at: <https://docs.web3j.io/>